

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Award		Centre Number 	Candidate Number
Time 3 hours		Paper reference	9811/01
Advanced Extension Award Mathematics			
You must have: Mathematical Formulae and Statistical Tables An insert for questions 5, 6 and 7			Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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1.

$$f(x) = x^{(x^2)} \quad x > 0$$

Use logarithms to find the x coordinate of the stationary point of the curve with equation $y = f(x)$.

(5)

$$y = x^{x^2}$$

$$\ln y = \ln x^{x^2}$$

$$\ln y = x^2 \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = 2x \ln x + x^2 \cdot \frac{1}{x}$$

$$\frac{1}{y} \frac{dy}{dx} = 2x \ln x + x$$

$$\frac{dy}{dx} = y (2x \ln x + x)$$

$$\frac{dy}{dx} = x^{x^2} (2x \ln x + x)$$

$$\frac{dy}{dx} = 0 : x^{x^2} (2x \ln x + x) = 0$$

$$x^{x^2} \cdot x \cdot (2 \ln x + 1) = 0$$

$$x^{x^2} \neq 0, x \neq 0, 2 \ln x + 1 = 0 \because x > 0$$

$$\ln x = -\frac{1}{2}$$

$$x = e^{-\frac{1}{2}}$$

$$\therefore x = \frac{1}{\sqrt{e}} \text{ ONLY}$$

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2.

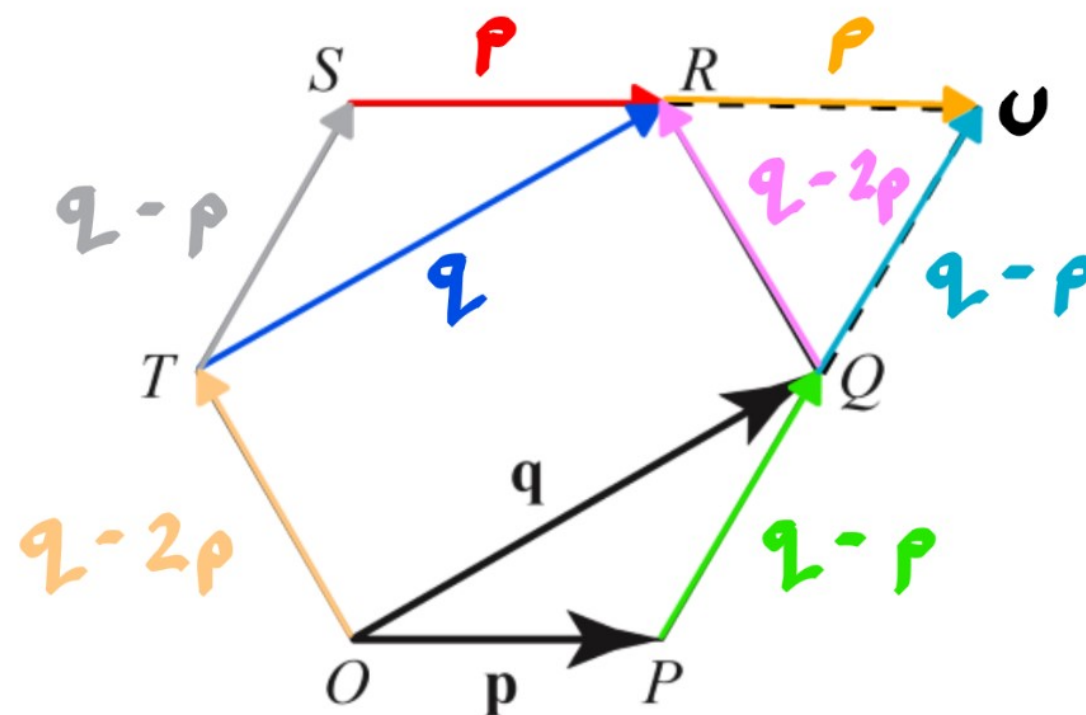


Figure 1

Figure 1 shows a regular hexagon $OPQRST$.

The vectors $\mathbf{p}$ and $\mathbf{q}$ are defined by $\mathbf{p} = \overrightarrow{OP}$ and $\mathbf{q} = \overrightarrow{OQ}$

Forces, in Newtons, $\mathbf{F}_P = (\overrightarrow{OP})$, $\mathbf{F}_Q = 2 \times (\overrightarrow{OQ})$, $\mathbf{F}_R = 3 \times (\overrightarrow{OR})$, $\mathbf{F}_S = 4 \times (\overrightarrow{OS})$ and $\mathbf{F}_T = 5 \times (\overrightarrow{OT})$ are applied to a particle.

(a) Find, in terms of $\mathbf{p}$ and $\mathbf{q}$, the resultant force on the particle.

(5)

The magnitude of the acceleration of the particle due to these forces is 13 ms^{-2}

Given that the mass of the particle is 3 kg,

(b) find $|\mathbf{p}|$

(5)

(a) $\overrightarrow{SR} = \overrightarrow{OP} = \mathbf{p}$

$\overrightarrow{TR} = \overrightarrow{OQ} = \mathbf{q}$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \mathbf{q} - \mathbf{p}$

$\overrightarrow{TS} = \overrightarrow{PQ} = \mathbf{q} - \mathbf{p}$

$\overrightarrow{RU} = \overrightarrow{OP} = \mathbf{p}$

$\overrightarrow{QU} = \overrightarrow{PQ} = \mathbf{q} - \mathbf{p}$

NOTE : WE HAVE EXTENDED THE LINES PQ AND SR TO A COMMON POINT U

$\overrightarrow{QR} = \overrightarrow{QU} + \overrightarrow{UR} = (\mathbf{q} - \mathbf{p}) + (-\mathbf{p}) = \mathbf{q} - 2\mathbf{p}$



Question 2 continued

$$\vec{OT} = \vec{QR} = q - 2p$$

$$\vec{OR} = \vec{OQ} + \vec{QR} = q + (q - 2p) = 2q - 2p$$

$$\vec{OS} = \vec{OR} + \vec{RS} = (2q - 2p) + (-p) = 2q - 3p$$

$$\therefore \vec{R} = \vec{F}_p + \vec{F}_q + \vec{F}_R + \vec{F}_S + \vec{F}_T$$

$$= \vec{OP} + 2\vec{OQ} + 3\vec{OR} + 4\vec{OS} + 5\vec{OT}$$

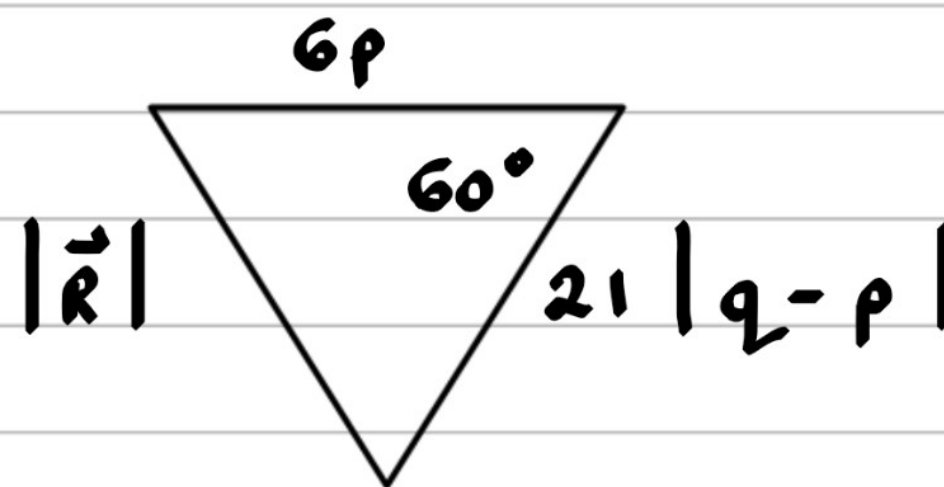
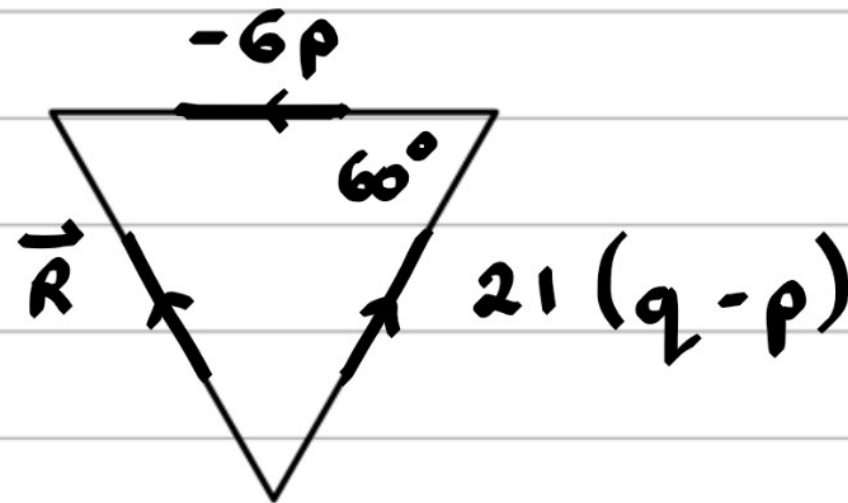
$$= p + 2q + 3(2q - 2p) + 4(2q - 3p) + 5(q - 2p)$$

$$= p + 2q + 6q - 6p + 8q - 12p + 5q - 10p$$

$$= -27p + 21q$$

(6) NEED TO FIND $|\vec{R}|$ IN TERMS OF p ONLY

$$\therefore \vec{R} = 21(q - p) - 6p$$

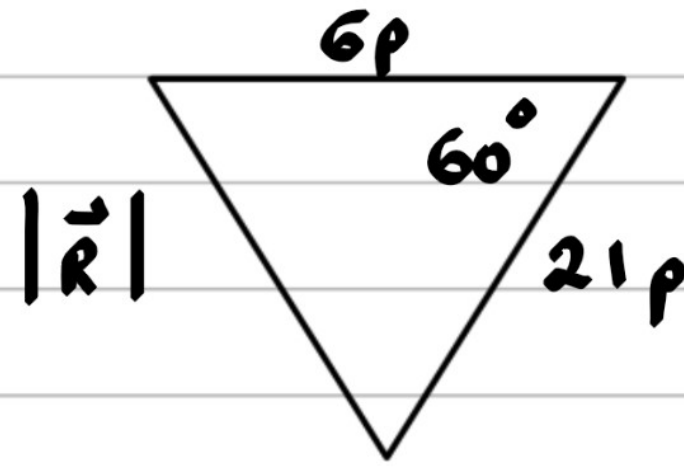


AS OPQRST IS A REGULAR HEXAGON:

$$\therefore |\vec{OP}| = |\vec{PQ}| \therefore |p| = |q - p| = p$$



Question 2 continued



$$\therefore |\vec{R}|^2 = (6p)^2 + (21p)^2 - 2(6p)(21p)\cos 60^\circ$$

$$= 36p^2 + 441p^2 - 252p^2 \times \frac{1}{2}$$

$$= 351p^2$$

$$\therefore |\vec{R}| = \sqrt{351p^2}$$

$$= 3p\sqrt{39}$$

$$\therefore \sum \text{FORCES} = ma : 3p\sqrt{39} = 3 \times 13$$

$$\therefore p = \frac{\sqrt{39}}{3}$$



3. (a) Use the formulae for $\sin(A \pm B)$ and $\cos(A \pm B)$ to prove that $\tan(90^\circ - \theta) \equiv \cot \theta$ (3)

(b) Solve for $0 < \theta < 360^\circ$

$$2 - \sec^2(\theta + 11^\circ) = 2 \tan(\theta + 11^\circ) \tan(\theta - 34^\circ)$$

Give each answer as an integer in degrees.

(8)

(+S1)

(a) LHS: $\tan(90^\circ - \theta) \equiv \frac{\sin(90^\circ - \theta)}{\cos(90^\circ - \theta)}$

$$\equiv \frac{\sin 90^\circ \cos \theta - \cos 90^\circ \sin \theta}{\cos 90^\circ \cos \theta + \sin 90^\circ \sin \theta}$$

$$\equiv \frac{\cos \theta - 0}{0 + \sin \theta}$$

$$\equiv \frac{\cos \theta}{\sin \theta}$$

$$\equiv \cot \theta$$

$$\equiv \text{RHS, HENCE SHOWN}$$

(b) $2 - \sec^2(\theta + 11) = 2 \tan(\theta + 11) \tan(\theta - 34)$, $0 < \theta < 360$

$$2 - [1 + \tan^2(\theta + 11)] = 2 \tan(\theta + 11) \tan(\theta - 34)$$

$$1 - \tan^2(\theta + 11) = 2 \tan(\theta + 11) \tan(\theta - 34)$$

$$\frac{1}{\tan(\theta - 34)} = \frac{2 \tan(\theta + 11)}{1 - \tan^2(\theta + 11)} \quad \text{NOTE: } \tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

$$\frac{1}{\tan(\theta - 34)} = \tan(2(\theta + 11))$$

$$\cot(\theta - 34) = \tan(2\theta + 22)$$



Question 3 continued

$$\tan(90 - (\theta - 34)) = \tan(2\theta + 22)$$

$$\tan(124 - \theta) = \tan(2\theta + 22)$$

$$124 - \theta = 2\theta + 22$$

$$3\theta = 102 + 180K, \quad K \in \mathbb{Z} \quad \text{NEW DOMAIN: } 0 < 3\theta < 1080$$

$$3\theta = 102, 282, 462, 642, 822, 1002$$

$$\therefore \theta = 34, 94, 154, 214, 274, 334$$



4. Given that $f(x) = e^{x^3-2x}$

(a) find $f'(x)$

(2)

The curves C_1 and C_2 are defined by the functions g and h respectively, where

$$g(x) = 8x^3e^{x^3-2x} \quad x \in \mathbb{R}, x > 0$$

$$h(x) = (3x^5 + 4x)e^{x^3-2x} \quad x \in \mathbb{R}, x > 0$$

(b) Find the x coordinates of the points of intersection of C_1 and C_2

(4)

Given that C_1 lies above C_2 between these points of intersection,

(c) find the area of the region bounded by the curves between these two points.

Give your answer in the form $A + Be^C$ where A , B , and C are exact real numbers to be found.

(7)

(+S1)

(a) $f(x) = e^{x^3-2x}$

$$f'(x) = (3x^2 - 2)e^{x^3-2x}$$

(b) $8x^3e^{x^3-2x} = (3x^5 + 4x)e^{x^3-2x}$

$$(3x^5 - 8x^3 + 4x)e^{x^3-2x} = 0$$

$$e^{x^3-2x} \neq 0, \quad 3x^5 - 8x^3 + 4x = 0$$

$$x(3x^4 - 8x^2 + 4) = 0$$

$$x(3x^2 - 2)(x^2 - 2) = 0$$

$$x = 0, \quad 3x^2 - 2 = 0, \quad x^2 - 2 = 0$$

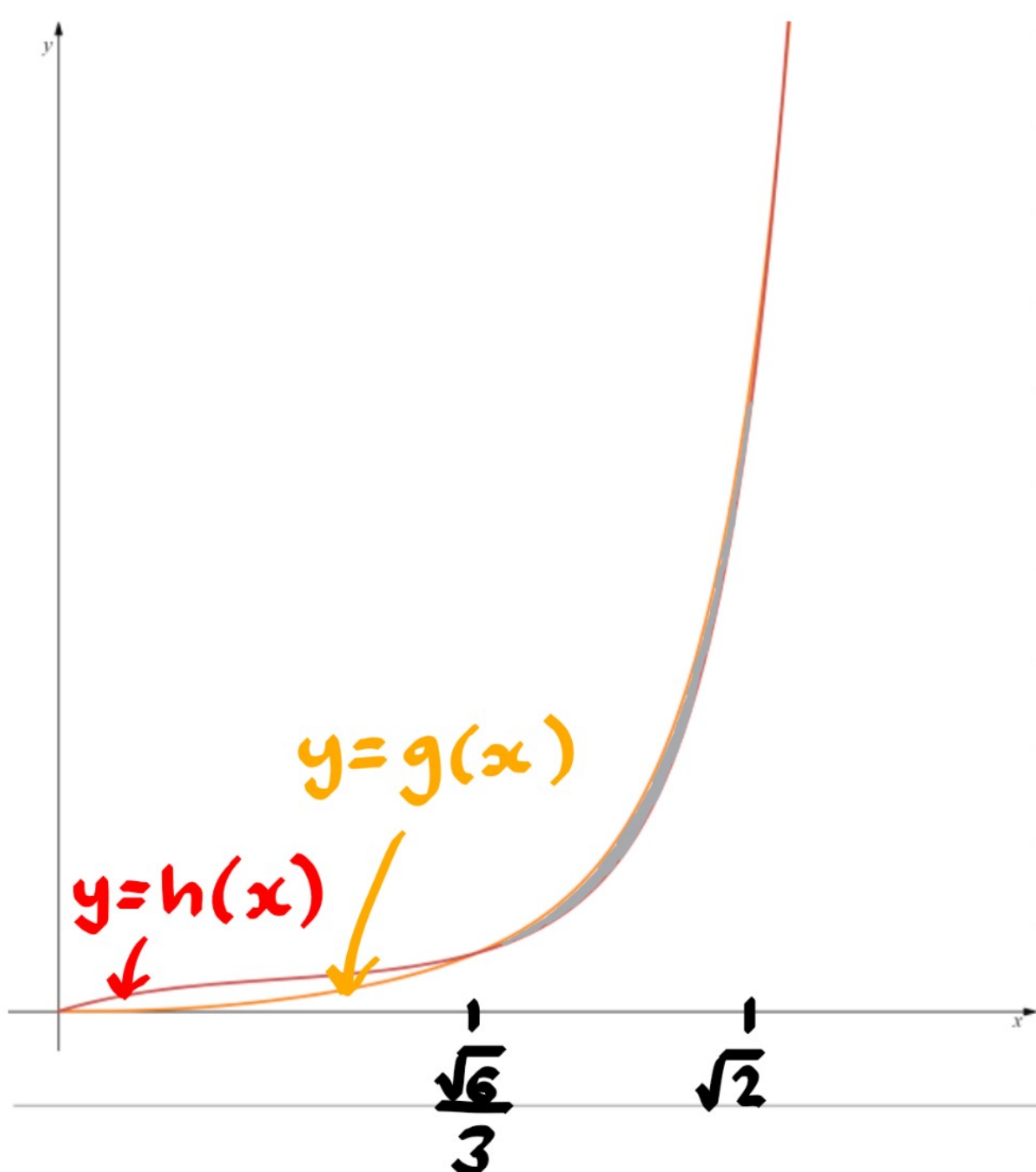
$$x = 0, \quad x = \pm \sqrt{\frac{2}{3}}, \quad x = \pm \sqrt{2}$$

$$\text{As } x > 0$$

$$\therefore x = \sqrt{\frac{2}{3}}, \sqrt{2} \quad \text{ONLY}$$



Question 4 continued



NOTE : $\sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{\sqrt{3}}$

$= \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{\sqrt{6}}{3}$

AREA = $\int_{\alpha}^{\beta} \{g(x) - h(x)\} dx$ WHERE : $\alpha = \frac{\sqrt{6}}{3}$, $\beta = \sqrt{2}$

$= \int_{\alpha}^{\beta} \{ 8x^3 e^{x^3-2x} - (3x^5 + 4x) e^{x^3-2x} \} dx$

$= \int_{\alpha}^{\beta} (8x^3 - 3x^5 - 4x) e^{x^3-2x} dx$

$= - \int_{\alpha}^{\beta} (3x^5 - 8x^3 + 4x) e^{x^3-2x} dx$

$= - \int_{\alpha}^{\beta} x (3x^4 - 8x^2 + 4) e^{x^3-2x} dx$

$= - \int_{\alpha}^{\beta} x (3x^2 - 2)(x^2 - 2) e^{x^3-2x} dx$

$= - \int_{\alpha}^{\beta} (x^3 - 2x)(3x^2 - 2) e^{x^3-2x} dx$



Question 4 continued

USE INTEGRATION BY PARTS:

$$u = x^3 - 2x$$

$$v = e^{x^3 - 2x}$$

$$u' = 3x^2 - 2$$

$$v' = (3x^2 - 2)e^{x^3 - 2x}$$

$$\text{AREA} = - \left\{ \left[(x^3 - 2x)e^{x^3 - 2x} \right]_{\alpha}^{\beta} - \int_{\alpha}^{\beta} (3x^2 - 2)e^{x^3 - 2x} dx \right\}$$

$$= - \left\{ \left[(x^3 - 2x)e^{x^3 - 2x} \right]_{\alpha}^{\beta} - \left[e^{x^3 - 2x} \right]_{\alpha}^{\beta} \right\}$$

$$= \left[- (x^3 - 2x)e^{x^3 - 2x} + e^{x^3 - 2x} \right]_{\alpha}^{\beta}$$

$$= \left[e^{x^3 - 2x} \{ 1 - (x^3 - 2x) \} \right]_{\alpha}^{\beta}$$

$$= \left[e^{x^3 - 2x} (1 - x^3 + 2x) \right]_{\alpha}^{\beta}$$

$$= \left\{ e^{\sqrt{2}^3 - 2\sqrt{2}} (1 - \sqrt{2}^3 + 2\sqrt{2}) \right\} - \left\{ e^{(\frac{\sqrt{6}}{3})^3 - 2(\frac{\sqrt{6}}{3})} \times (1 - (\frac{\sqrt{6}}{3})^3 + 2(\frac{\sqrt{6}}{3})) \right\}$$

$$= \left\{ e^{2\sqrt{2} - 2\sqrt{2}} (1 - 2\sqrt{2} + 2\sqrt{2}) \right\} - \left\{ e^{\frac{6\sqrt{6}}{27} - \frac{2\sqrt{6}}{3}} \times (1 - \frac{6\sqrt{6}}{27} + \frac{2\sqrt{6}}{3}) \right\}$$

$$= e^0 (1) - \left\{ e^{-\frac{4\sqrt{6}}{9}} (1 - \frac{4\sqrt{6}}{9}) \right\}$$

$$= 1 - \frac{4\sqrt{6} + 9}{9} e^{-\frac{4\sqrt{6}}{9}}$$

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5. An aeroplane leaves a runway and moves with a constant speed of V km/h due north along a straight path inclined at an angle $\arctan\left(\frac{3}{4}\right)$ to the horizontal.

A light aircraft is moving due north in a straight horizontal line in the same vertical plane as the aeroplane, at a height of 3 km above the runway.

The light aircraft is travelling with a constant speed of $2V$ km/h.

At the moment the aeroplane leaves the runway, the light aircraft is at a horizontal distance d km behind the aeroplane.

Both aircraft continue to move with the same trajectories due north.

- (a) Show that the distance, D km, between the two aircraft t hours after the aeroplane leaves the runway satisfies

$$D^2 = \left(\frac{6}{5}Vt - d\right)^2 + \left(\frac{3}{5}Vt - 3\right)^2 \quad (3)$$

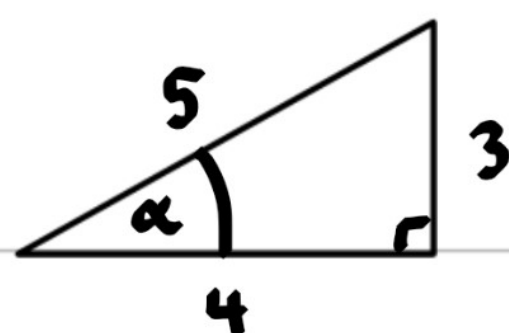
Given that the distance between the two aircraft is never less than 2 km,

- (b) find the range of possible values for d .

(7)

(+S1)

(a)



$$\sin \alpha = \frac{3}{5}, \cos \alpha = \frac{4}{5}, \tan \alpha = \frac{3}{4}$$

$$\text{AIRCRAFT } (2Vt - d, 3) \quad \text{AEROPLANE } (Vt \cos \alpha, Vt \sin \alpha) = \left(\frac{4}{5}Vt, \frac{3}{5}Vt\right)$$



$$D^2 = \left[\frac{4}{5}Vt - (2Vt - d)\right]^2 + \left[\frac{3}{5}Vt - 3\right]^2$$

$$= \left[\frac{4}{5}Vt - 2Vt + d\right]^2 + \left[\frac{3}{5}Vt - 3\right]^2$$

$$= \left[-\frac{6}{5}Vt + d\right]^2 + \left[\frac{3}{5}Vt - 3\right]^2$$

$$= (-1)^2 \left[\frac{6}{5}Vt - d\right]^2 + \left[\frac{3}{5}Vt - 3\right]^2$$

$$= \left[\frac{6}{5}Vt - d\right]^2 + \left[\frac{3}{5}Vt - 3\right]^2$$



Question 5 continued

(6) $D \geq 2 \therefore D^2 \geq 4$

$$\left[\frac{6vt}{5} - d \right]^2 + \left[\frac{3vt}{5} - 3 \right]^2 \geq 4$$

$$\frac{36v^2t^2}{25} - \frac{12vtd}{5} + d^2 + \frac{9v^2t^2}{25} - \frac{18vt}{5} + 9 \geq 4$$

$$\frac{9}{5} (vt)^2 - \left(\frac{12d+18}{5} \right) vt + d^2 + 5 \geq 0$$

NOTE: AS QUADRATIC IS POSITIVE, IT CANNOT HAVE ANY SOLUTIONS WHEN YOU SET IT EQUAL TO ZERO

$$b^2 - 4ac < 0: \left\{ - \left(\frac{12d+18}{5} \right) \right\}^2 - 4 \left(\frac{9}{5} \right) (d^2 + 5) < 0$$

$$\left\{ \frac{6(2d+3)}{5} \right\}^2 - \frac{36}{5} (d^2 + 5) < 0$$

$$\frac{36}{25} (2d+3)^2 - \frac{36}{5} (d^2 + 5) < 0$$

$$(2d+3)^2 - 5(d^2 + 5) < 0$$

$$4d^2 + 12d + 9 - 5d^2 - 25 < 0$$

$$-d^2 + 12d - 16 < 0$$

$$d^2 - 12d + 16 > 0$$

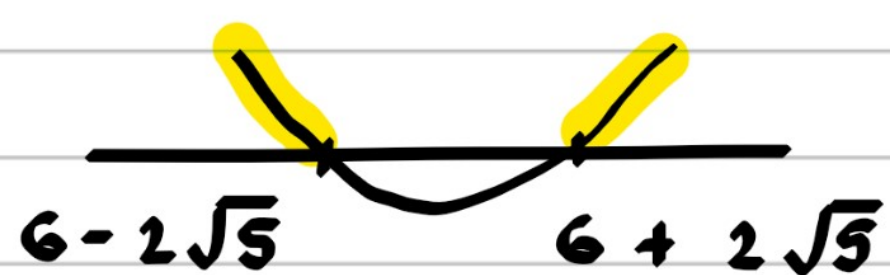
$$(d-6)^2 - 36 + 16 > 0$$

$$(d-6)^2 > 20$$



Question 5 continued

$$CV: d = 6 \pm 2\sqrt{5}$$



$$\therefore d > 6 + 2\sqrt{5}, d < 6 - 2\sqrt{5}$$

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6.

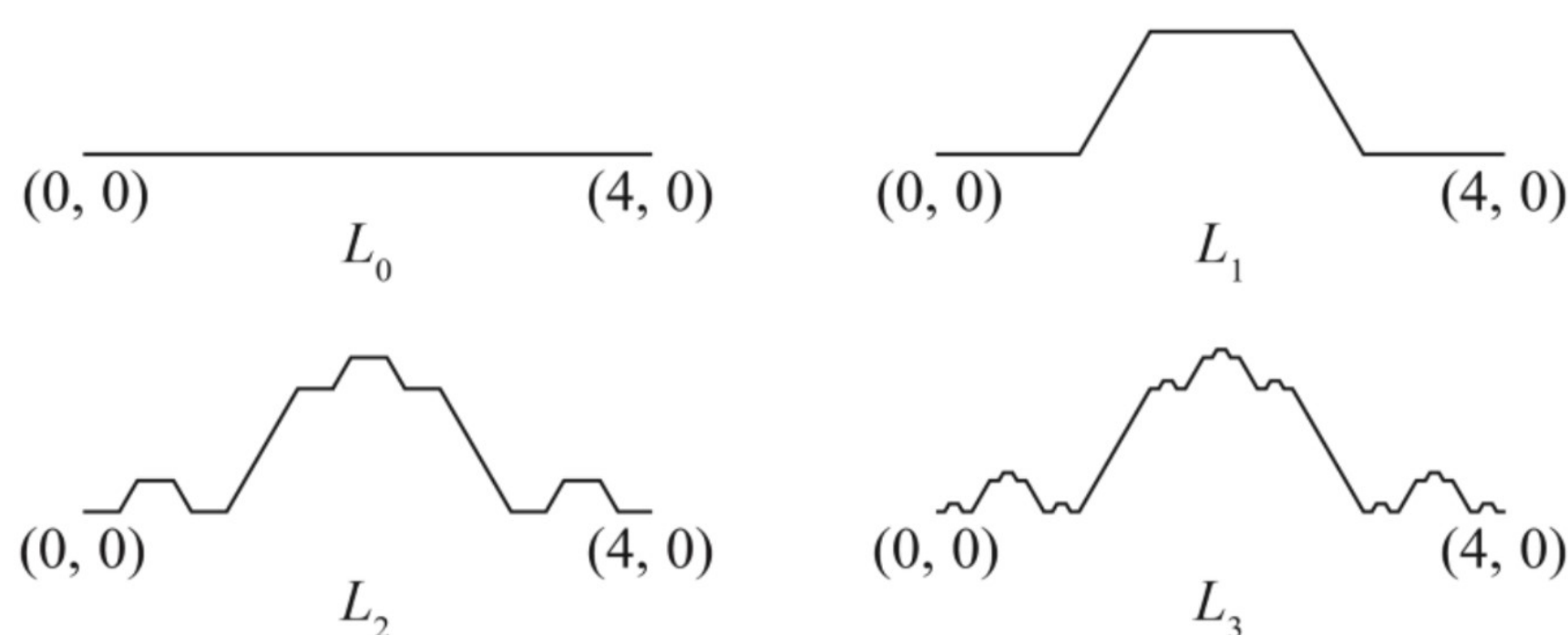


Figure 2

Figure 2 shows the first few iterations in the construction of a curve, L .

Starting with a straight line L_0 of length 4, the middle half of this line is replaced by three sides of a trapezium above L_0 as shown, such that the length of each of these sides is $\frac{1}{4}$ of the length of L_0 .

After the first iteration each line segment has length one.

In subsequent iterations, each line segment parallel to L_0 similarly has its middle half replaced by three sides of a trapezium above that line segment, with each side $\frac{1}{4}$ the length of that line segment.

Line segments in L_n are either parallel to L_0 or are sloped.

(a) Show that the length of L_2 is $\frac{23}{4}$ (2)

(b) Write down the number of

- (i) line segments in L_n that are parallel to L_0
- (ii) sloped line segments in L_2 that are not in L_1
- (iii) new sloped line segments that are created by the $(n + 1)$ th iteration. (3)

(c) Hence find the length of L_n as $n \rightarrow \infty$ (6)

The area enclosed between L_0 and L_n is A_n

(d) Find the value of A_1 (2)

(e) Find, in terms of n , an expression for $A_{n+1} - A_n$ (3)

(f) Hence find the value of A_n as $n \rightarrow \infty$ (3)

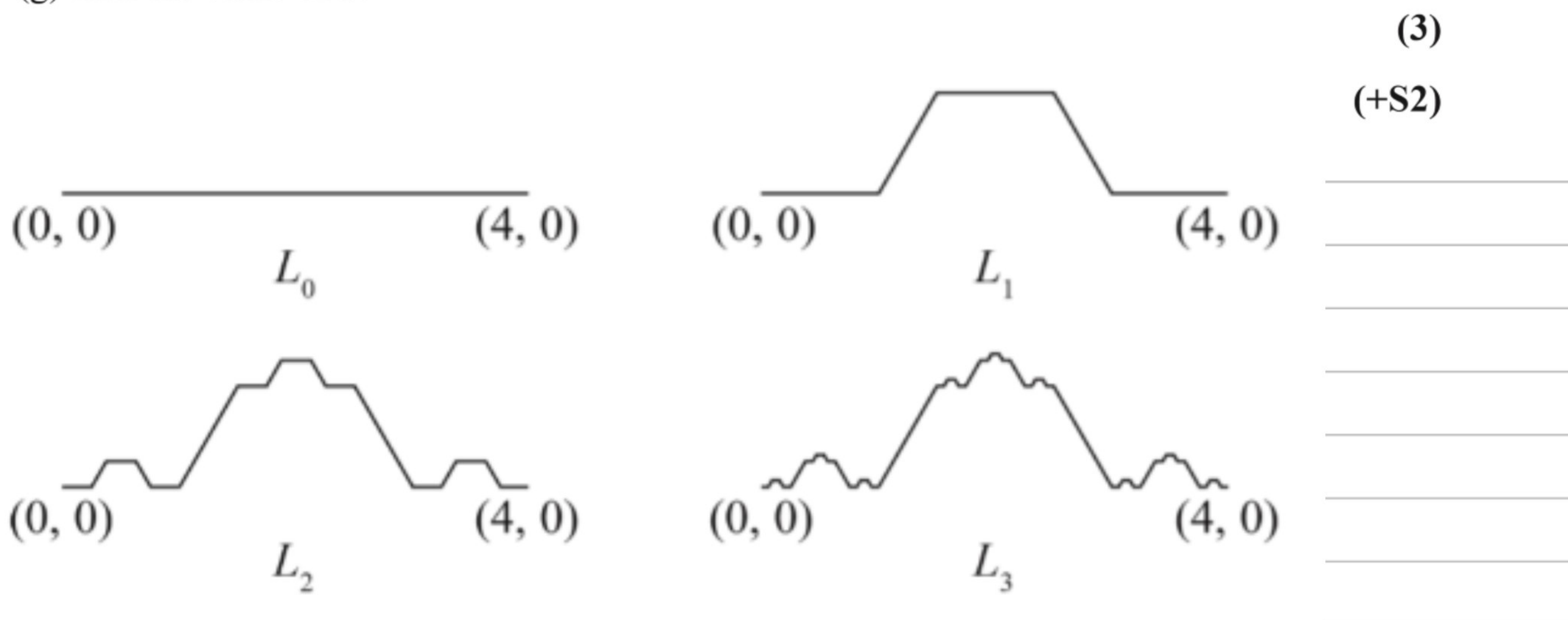


Question 6 continued

The same construction as described above is applied externally to the three sides of an equilateral triangle of side length a .

Given that the limit of the area of the resulting shape is $26\sqrt{3}$

(g) find the value of a .



$$L_0 = 4$$

$$L_1 = 5 \times \left(\frac{1}{4} \times 4\right) = 5$$

$$L_2 = 2 \times \left(\frac{1}{4} \times 4\right) + \frac{1}{4} \times \left(\frac{1}{4} \times 4\right) \times 5 \times 3 = \frac{23}{4}$$

L_1 HAS $3 = 3^1$ HORIZONTAL LINE SEGMENTS PARALLEL TO L_0

L_2 HAS $9 = 3^2$ HORIZONTAL LINE SEGMENTS PARALLEL TO L_0

L_3 HAS $27 = 3^3$ HORIZONTAL LINE SEGMENTS PARALLEL TO L_0

$\vdots$

L_n HAS 3^n HORIZONTAL LINE SEGMENTS PARALLEL TO L_0

EACH HORIZONTAL LINE SPLITS TO 3 NEW HORIZONTAL LINES AND 2 NEW SLOPED LINES

$\therefore 3 \times 2 = 6$ SLOPED LINE SEGMENTS IN L_2 THAT ARE NOT IN L_1



Question 6 continued

(iii) EACH HORIZONTAL LINE SPLITS INTO 2 NEW SLOPED LINES

$\therefore 2 \times 3^n$ NEW SLOPED LINE SEGMENTS THAT ARE CREATED BY THE $(n+1)$ TH ITERATION

(b) LENGTH OF SLOPES:

$$S_0 = 0$$

$$S_1 = 0 + 2 \times 1$$

$$= \left(\frac{3}{4}\right)^0 \times 2$$

$$S_2 = 0 + 2 \times 1 + 3 \times 2 \times \frac{1}{4} \times 1$$

$$= \left(\frac{3}{4}\right)^0 \times 2 + \left(\frac{3}{4}\right)^1 \times 2$$

$$S_3 = 0 + 2 \times 1 + 3 \times 2 \times \frac{1}{4} \times 1 + 3 \times \frac{1}{4} \times (3 \times 2 \times \frac{1}{4} \times 1)$$

$$= 0 + 2 + 3^1 \times 2 \times \left(\frac{1}{4}\right)^1 + 3^2 \times \left(\frac{1}{4}\right)^2 \times 2$$

$$= 0 + \left(\frac{3}{4}\right)^0 \times 2 + \left(\frac{3}{4}\right)^1 \times 2 + \left(\frac{3}{4}\right)^2 \times 2$$

$$S_4 = \left(\frac{3}{4}\right)^0 \times 2 + \left(\frac{3}{4}\right)^1 \times 2 + \left(\frac{3}{4}\right)^2 \times 2 + \left(\frac{3}{4}\right)^3 \times 2$$

$\vdots$

$\vdots$

$$S_{n+1} = \left(\frac{3}{4}\right)^0 \times 2 + \left(\frac{3}{4}\right)^1 \times 2 + \left(\frac{3}{4}\right)^2 \times 2 + \left(\frac{3}{4}\right)^3 \times 2 + \dots + \left(\frac{3}{4}\right)^n \times 2$$

$$= 2 \sum_{r=0}^n \left(\frac{3}{4}\right)^r$$

LENGTH OF HORIZONTALS

$$H_0 = 4$$

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Question 6 continued

$$H_1 = 3 \times \left(\frac{1}{4} \times 4\right)$$

$$H_2 = 3^2 \times \frac{1}{4} \left(\frac{1}{4} \times 4\right)$$

$$= 3^2 \times \left(\frac{1}{4}\right)^2 \times 4$$

$$= \left(\frac{3}{4}\right)^2 \times 4$$

$$H_3 = 3^3 \times \frac{1}{4} \left(\frac{1}{4} \times \frac{1}{4} \times 4\right)$$

$$= 3^3 \times \left(\frac{1}{4}\right)^3 \times 4$$

$$= \left(\frac{3}{4}\right)^3 \times 4$$

$$H_4 = \left(\frac{3}{4}\right)^4 \times 4$$

$$\vdots$$

$$H_{n+1} = \left(\frac{3}{4}\right)^{n+1} \times 4$$

$$= \frac{3^{n+1}}{4^{n+1}} \times 4$$

$$= 3^{n+1} \times \frac{4}{4^{n+1}}$$

$$= 3^{n+1} \times \left(\frac{1}{4}\right)^n$$

$$= 3^1 \times 3^n \times \left(\frac{1}{4}\right)^n$$

$$= 3 \left(\frac{3}{4}\right)^n$$

$$\therefore \text{TOTAL LENGTH, } L_{n+1} = 2 \sum_{r=0}^n \left(\frac{3}{4}\right)^r + 3 \left(\frac{3}{4}\right)^n$$



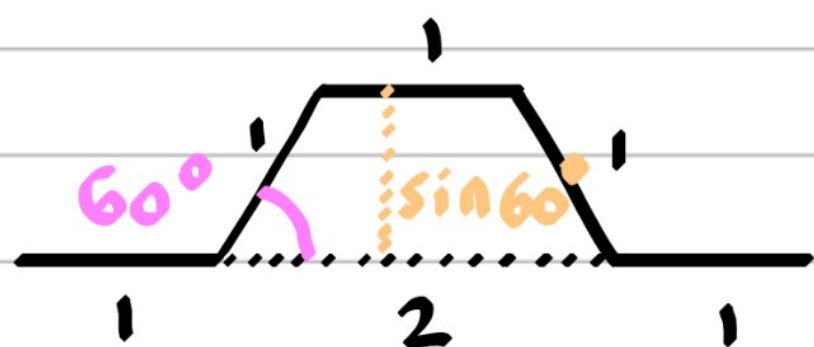
Question 6 continued

$$\text{AS } n \rightarrow \infty : 2 \sum_{r=0}^{\infty} \left(\frac{3}{4}\right)^r = 2 \times \frac{1}{1 - \frac{3}{4}} = 8$$

$$3 \left(\frac{3}{4}\right)^n \rightarrow 0$$

$$\therefore L_{\infty} \rightarrow 8$$

(d)



$$\begin{aligned} \text{AREA, } A_1 &= \frac{(2+1) \sin 60^\circ}{2} \\ &= \frac{3\sqrt{3}}{4} \end{aligned}$$

(e) LENGTH SCALE FACTOR IS $\frac{1}{4}$

$$\therefore \text{AREA SCALE FACTOR IS } \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

SO FROM FIRST TO SECOND WE HAVE 3 NEW TRAPEZIA $\left(\frac{1}{16}\right)^{\text{TH}}$ THE AREA OF THE FIRST ONE IN THE FIRST ITERATION

IN THE $(n+1)^{\text{TH}}$ ITERATION THE NEW TRAPEZIA HAVE AREA $\left(\frac{1}{16}\right)^n$ AND THERE ARE 3^n SUCH

$$\therefore \text{AREA INCREASES BY } 3^n \times \frac{3\sqrt{3}}{4} \times \left(\frac{1}{16}\right)^n$$

$$(f) A_{\infty} = A_1 + A_2 + A_3 + A_4 + \dots$$

$$= \frac{3\sqrt{3}}{4} + 3^1 \times \frac{1}{16^1} \times \frac{3\sqrt{3}}{4} + 3^2 \times \frac{1}{16^2} \times \frac{3\sqrt{3}}{4} + 3^3 \times \frac{1}{16^3} \times \frac{3\sqrt{3}}{4} + \dots$$



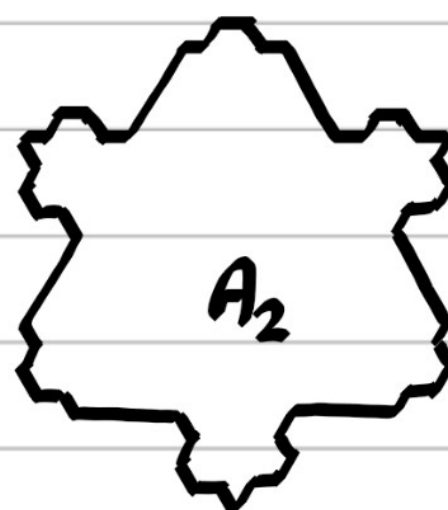
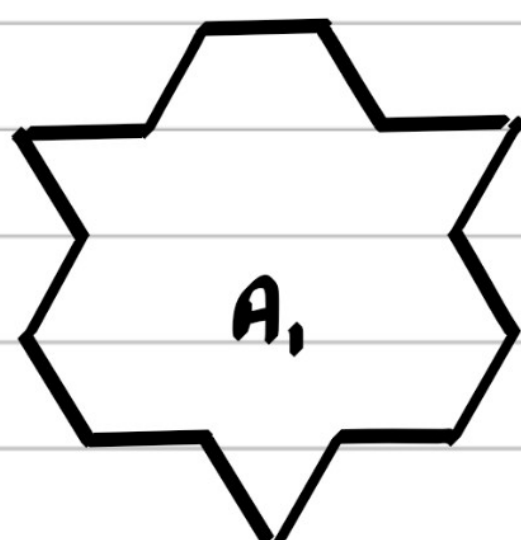
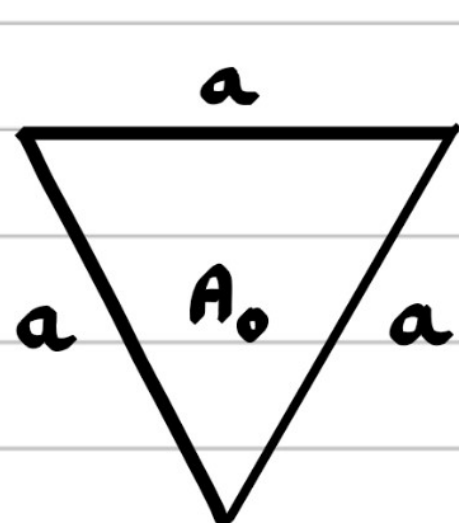
Question 6 continued

$$= \frac{3\sqrt{3}}{4} + \left(\frac{3}{16}\right)^1 \times \frac{3\sqrt{3}}{4} + \left(\frac{3}{16}\right)^2 \times \frac{3\sqrt{3}}{4} + \left(\frac{3}{16}\right)^3 \times \frac{3\sqrt{3}}{4} + \dots$$

$$= \frac{\frac{3\sqrt{3}}{4}}{1 - \frac{3}{16}}$$

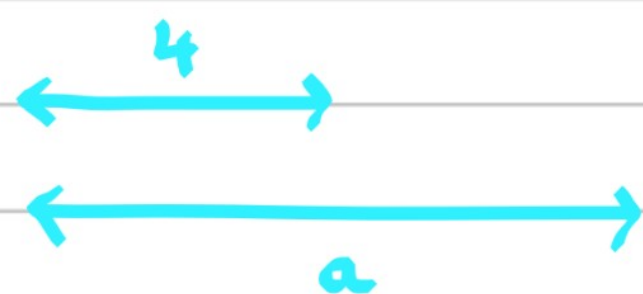
$$= \frac{3\sqrt{3}}{4} \cdot \frac{16}{13}$$

$$= \frac{12\sqrt{3}}{13}$$



$$\text{AREA OF TRIANGLE} = \frac{1}{2} \times a^2 \times \sin 60$$

$$= \frac{a^2 \sqrt{3}}{4}$$



$$\therefore \text{LENGTH SCALE FACTOR} = \frac{a}{4}$$

$$\therefore \text{AREA SCALE FACTOR} = \left(\frac{a}{4}\right)^2$$



Question 6 continued

LIMITING AREA OF SHAPE:

$$\frac{a^2 \sqrt{3}}{4} + 3 \left(\frac{a}{4} \right)^2 \times \frac{12 \sqrt{3}}{13} = 26 \sqrt{3}$$

$$\frac{a^2}{4} + \frac{3a^2}{16} \cdot \frac{12}{13} = 26$$

$$a^2 \left\{ \frac{1}{4} + \frac{3}{16} \cdot \frac{12}{13} \right\} = 26$$

$$a^2 \left\{ \frac{1}{4} + \frac{3 \times 3}{4 \times 13} \right\} = 26$$

$$a^2 \left\{ \frac{1}{4} + \frac{9}{52} \right\} = 26$$

$$a^2 \left\{ \frac{13 + 9}{52} \right\} = 26$$

$$\frac{11a^2}{52} = 26$$

$$a^2 = \frac{26^2}{11}$$

$$\therefore a = \frac{26}{\sqrt{11}}$$

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7. A circle C has centre $X(a, b)$ and radius r .

A line l has equation $y = mx + c$

(a) Show that the x coordinates of the points where C and l intersect satisfy

$$(m^2 + 1)x^2 - 2(a - m(c - b))x + a^2 + (c - b)^2 - r^2 = 0 \quad (2)$$

Given that l is a tangent to C ,

(b) show that

$$c = b - ma \pm r\sqrt{m^2 + 1} \quad (6)$$

The circle C_1 has equation

$$x^2 + y^2 - 16 = 0$$

and the circle C_2 has equation

$$x^2 + y^2 - 20x - 10y + 89 = 0$$

(c) Find the equations of any lines that are normal to both C_1 and C_2 , justifying your answer.

(3)

(d) Find the equations of all lines that are a tangent to both C_1 and C_2

[You may find the following Pythagorean triple helpful in this part: $7^2 + 24^2 = 25^2$]
 $\therefore 25^2 - 7^2 = 24^2$ (11)

(+S2)

(a) $C: (x - a)^2 + (y - b)^2 = r^2$ ①

$l: y = mx + c$ ②

② $\rightarrow$ ①: $(x - a)^2 + (mx + (c - b))^2 = r^2$

$(x - a)(x - a) + (mx + (c - b))(mx + (c - b)) = r^2$

$x^2 - 2ax + a^2 + m^2x^2 + 2mx(c - b) + (c - b)^2 - r^2 = 0$

$(m^2 + 1)x^2 - 2[a - m(c - b)]x + a^2 + (c - b)^2 - r^2 = 0$

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Question 7 continued

(6)

$$a = m^2 + 1, b = -2[a - m(c - b)], c = a^2 + (c - b)^2 - r^2$$

$$b^2 - 4ac = 0: \{-2[a - m(c - b)]\}^2 - 4(m^2 + 1)\{a^2 + (c - b)^2 - r^2\} = 0$$

$$\Rightarrow 4[a - m(c - b)]^2 - (4m^2 + 4)[a^2 + (c - b)^2 - r^2] = 0$$

$$\Rightarrow 4[a^2 - 2am(c - b) + m^2(c - b)^2] - (4m^2 + 4) \times [a^2 + (c - b)^2 - r^2] = 0$$

$$\Rightarrow 4a^2 - 8am(c - b) + 4m^2(c - b)^2 - 4m^2a^2 - 4m^2(c - b)^2 + 4m^2r^2 - 4[a^2 + (c - b)^2 - r^2] = 0$$

$$\Rightarrow \cancel{a^2} - 2am(c - b) + \cancel{m^2(c - b)^2} - \cancel{m^2a^2} - \cancel{m^2(c - b)^2} + m^2r^2 - \cancel{a^2} - (c - b)^2 + r^2 = 0$$

$$\Rightarrow -2am(c - b) - m^2a^2 + m^2r^2 - (c - b)^2 + r^2 = 0$$

$$\Rightarrow (c - b)^2 + 2am(c - b) + m^2a^2 - m^2r^2 - r^2 = 0$$

LET $\lambda = c - b$:

$$\Rightarrow \lambda^2 + 2am\lambda + m^2a^2 - m^2r^2 - r^2 = 0$$

$$\Rightarrow (\lambda + am)^2 - \cancel{a^2m^2} + \cancel{m^2a^2} - m^2r^2 - r^2 = 0$$

$$\Rightarrow (\lambda + am)^2 = m^2r^2 + r^2$$

$$\Rightarrow (c - b + am)^2 = m^2r^2 + r^2$$

$$\Rightarrow (c - b + am)^2 = r^2(m^2 + 1)$$

$$\Rightarrow c - b + am = \pm \sqrt{r^2(m^2 + 1)}$$



Question 7 continued

$$\Rightarrow c - b + am = \pm r \sqrt{m^2 + 1}$$

$$\therefore c = b - am \pm r \sqrt{m^2 + 1}$$

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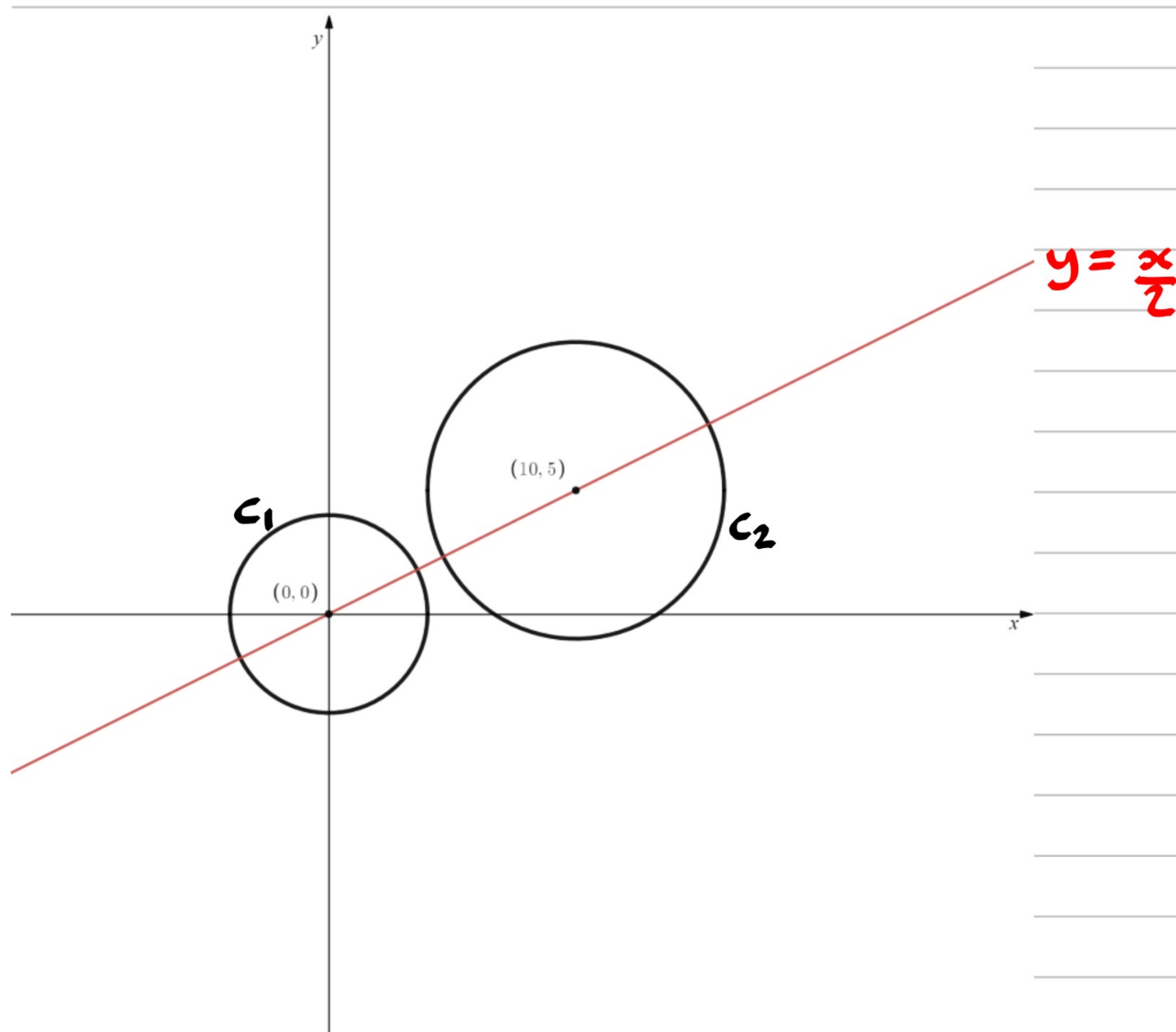
(c) $C_1: x^2 + y^2 = 16$, CENTRE $(0, 0)$, RADIUS = 4

$$C_2: x^2 + y^2 - 20x - 10y + 89 = 0$$

$$(x - 10)^2 + (y - 5)^2 - 100 - 25 + 89 = 0$$

$$(x - 10)^2 + (y - 5)^2 = 36, \text{ CENTRE } (10, 5), \text{ RADIUS} = 6$$

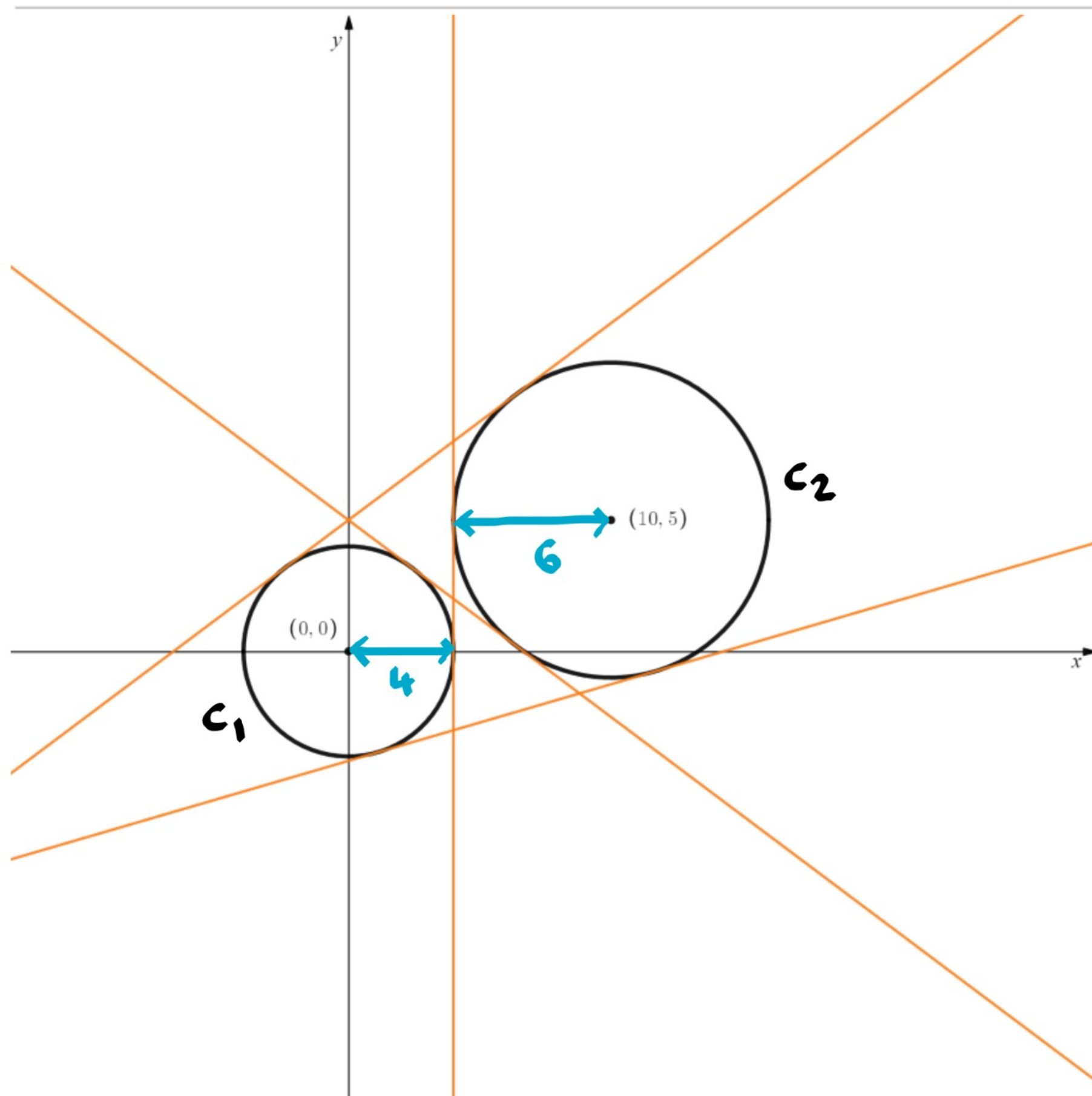
NOTE: ALL NORMALS TO A CIRCLE MUST PASS THROUGH THEIR CENTRES, SO THERE IS ONLY ONE COMMON NORMAL, THE LINE THROUGH BOTH CENTRES



Question 7 continued

$$\text{GRADIENT} = \frac{5}{10} = \frac{1}{2}$$

$\therefore$ NORMAL TO BOTH C_1 AND C_2 HAS EQUATION $y = \frac{1}{2}x$



HORIZONTAL DISTANCE BETWEEN BOTH CENTRES IS 10 AND THE SUM OF THE RADII IS $4 + 6 = 10$

$\therefore$ THERE IS A VERTICAL TANGENT AT $x = 4$

USING (b), OTHER COMMON TANGENTS $y = mx + c$ SATISFY:

$$c = \underbrace{0 - 0m \pm 4\sqrt{m^2 + 1}}_{C_1} = \underbrace{5 - 10m \pm 6\sqrt{m^2 + 1}}_{C_2}$$

$$c = \pm 4\sqrt{m^2 + 1} = 5 - 10m \pm 6\sqrt{m^2 + 1}$$



Question 7 continued

$$\pm 10 \sqrt{m^2 + 1} = 5 - 10m \text{ ③ AND } \pm 2 \sqrt{m^2 + 1} = 5 - 10m \text{ ④}$$

$$\text{FROM ③: } \pm 10 \sqrt{m^2 + 1} = 5 - 10m$$

$$100(m^2 + 1) = (5 - 10m)^2$$

$$100m^2 + 100 = 25 - 100m + 100m^2$$

$$100m = -75$$

$$\therefore m = -\frac{3}{4}$$

$$\text{FROM ④: } \pm 2 \sqrt{m^2 + 1} = 5 - 10m$$

$$4(m^2 + 1) = (5 - 10m)^2$$

$$4m^2 + 4 = 25 - 100m + 100m^2$$

$$96m^2 - 100m + 21 = 0$$

$$(24m - 7)(4m - 3) = 0$$

$$\therefore m = \frac{3}{4}, \frac{7}{24}$$

$$\text{BUT } c = \pm 4 \sqrt{m^2 + 1} :$$

$$m = \pm \frac{3}{4} : c = \pm 4 \sqrt{\left(\frac{3}{4}\right)^2 + 1}$$

$$c = \pm 4 \times \sqrt{\frac{25}{16}}$$

$$c = \pm 4 \times \frac{5}{4}$$

$$\therefore c = \pm 5$$

$$m = \frac{7}{24} : c = \pm 4 \sqrt{\left(\frac{7}{24}\right)^2 + 1}$$

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Question 7 continued

$$C = \pm 4 \sqrt{\frac{7^2}{24^2} + 1}$$

$$C = \pm 4 \sqrt{\frac{7^2}{24^2} + \frac{25^2 - 7^2}{24^2}}$$

$$C = \pm 4 \sqrt{\frac{25^2}{24^2}}$$

$$C = \pm 4 \times \frac{25}{24}$$

$$\therefore C = \pm \frac{25}{6}$$

$\therefore$ ALL OF THE EQUATIONS OF TANGENTS: $x = 4$

$$y = \frac{3}{4}x + 5$$

$$y = -\frac{3}{4}x + 5$$

$$y = \frac{7}{24}x - \frac{25}{6}$$

